

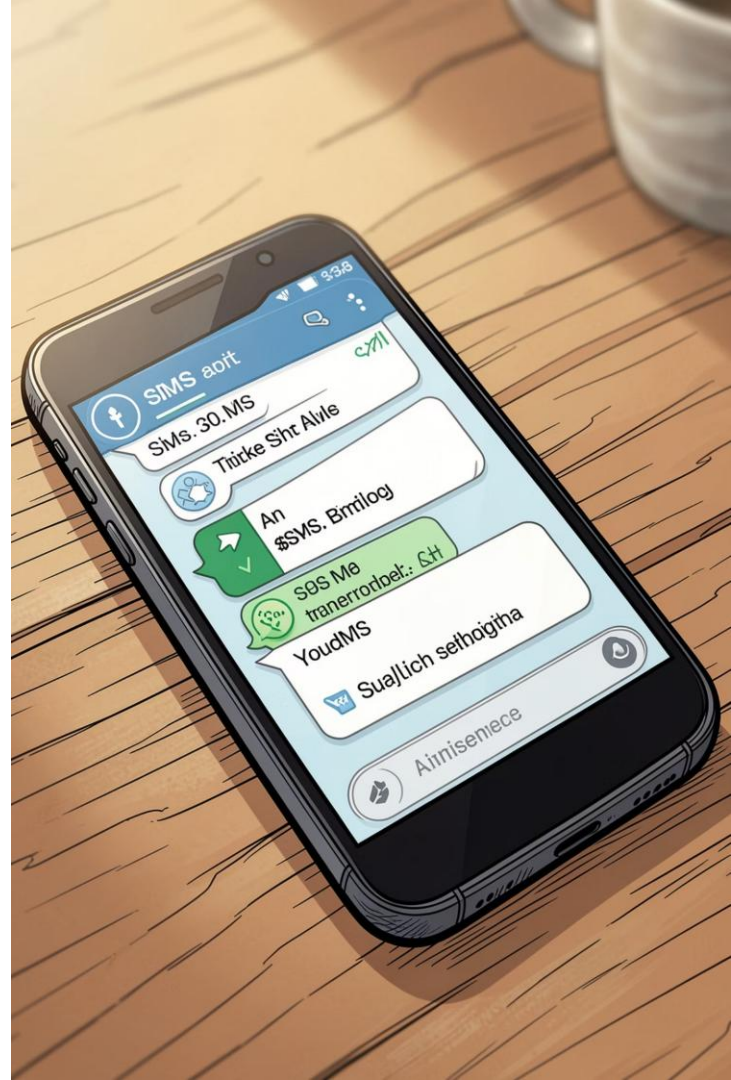
MobiScan via SMS

An Android mobile security and tracking application that operates entirely over SMS — enabling remote control and protection when internet connectivity is unavailable.

PADALINGAM S

DEPT. OF COMPUTER SCIENCE & ENGINEERING (CYBER SECURITY)

SRI SHAKTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY



Overview

MobiScan via SMS provides remote device control and tracking using SMS commands. It is designed for situations where internet-based solutions fail — for instance, when the device has no data, is in airplane mode with SMS allowed, or cannot sign into cloud services.



Remote Locking

Instantly lock a lost or stolen device using a secure SMS command.



GPS Tracking

Obtain coordinates and a Google Maps link via SMS — no internet required on the controller side.



Ring & Vibrate

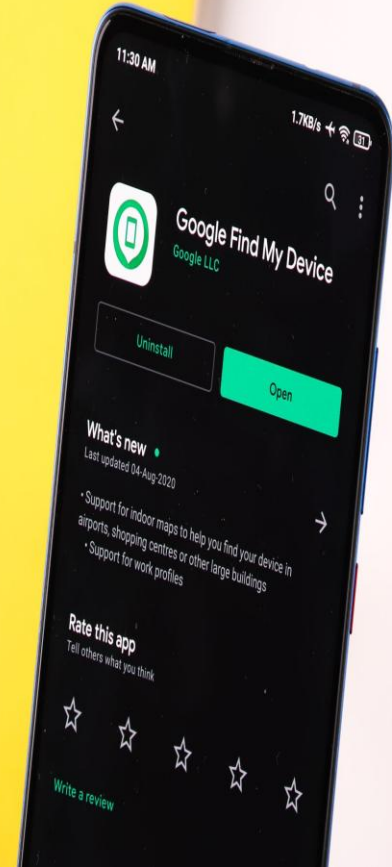
Trigger audible/vibration alerts even when the device is silenced.

Problem Statement

Many tracking solutions require continuous internet, cloud accounts, and online connectivity. When a device is offline or lacks a logged-in cloud session, traditional trackers become ineffective — leaving users without recourse.

Objectives

- Enable remote location tracking via SMS
- Allow device locking through SMS commands
- Provide audible alerts (ring/vibrate) remotely
- Detect and log SIM changes
- Operate reliably without internet



Technologies & Components



Platform

Android (Kotlin, XML)



Database

Firebase Realtime Database for event logging and SIM history



Messaging

SmsManager and BroadcastReceiver to receive/process commands



Location

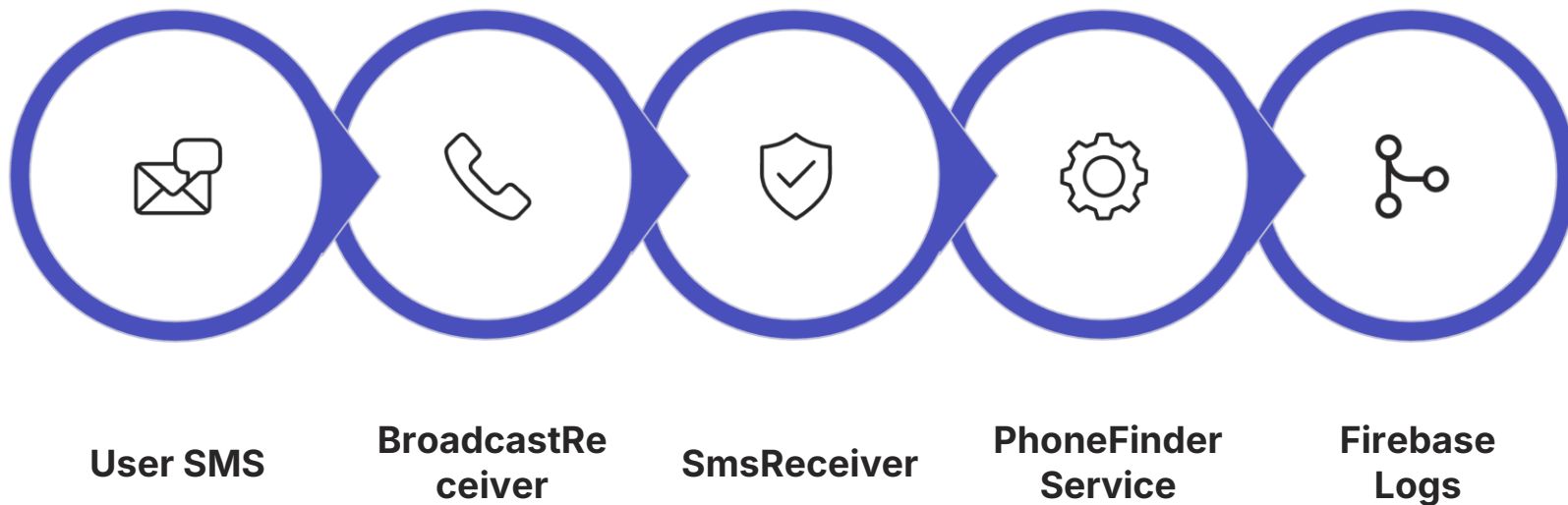
FusedLocationProviderClient for GPS coordinates

Key Android APIs: BroadcastReceiver, DeviceAdminReceiver, DevicePolicyManager, SmsManager, FusedLocationProviderClient.

Software Components

Component	Purpose
BroadcastReceiver	Receives incoming SMS messages and hands them to the processor
SmsReceiver	Parses and verifies SMS commands; authenticates sender if required
PhoneFinderService	Manages background tasks: fetches location, triggers ring/vibrate
DeviceAdminReceiver	Obtains admin permissions necessary to lock the device
SmsManager	Sends SMS responses with status or location links
Firebase Database	Stores SIM change logs, command audit trail, and device metadata

System Architecture — Workflow



This SMS-first architecture ensures operations complete even when the device lacks active internet — Firebase logging happens when connectivity resumes.

Working Process (Example: MOBI LOCK)



01

1. Send Command

User sends SMS: MOBI LOCK.

02

2. Receive & Verify

BroadcastReceiver captures SMS; SmsReceiver validates the format and sender.

03

3. Execute

DevicePolicyManager via DeviceAdminReceiver locks the device immediately.

04

4. Respond

SmsManager sends confirmation back to the requester; event logged to Firebase.

Commands & Features

MobiScan accepts a concise command set designed for reliability and simplicity. Each command triggers a focused action and returns a compact SMS response.



MOBI RING / MOBI BUZZ

Rings or vibrates the phone to locate it physically.



MOBI LOCATION / MOBI GPS

Returns GPS coordinates and a Google Maps link via SMS.



MOBI LOCK / MOBI SECURE

Locks the device instantly using DevicePolicyManager.



MOBI SIM

Reports current SIM details; detects and logs SIM changes to Firebase.



MOBI HELP

Sends the full list of available commands and usage tips.

Advantages & Future Scope

Offline Operation

Works when the device has no active data connection.

SMS Reliability

SMS is more resilient in low-connectivity and across networks.

Security Controls

Device admin features allow enforcement actions (lock, secure).

Audit Trail

Firebase stores SIM/log events for post-incident review.

Planned Enhancements

- Face recognition for verified unlock and owner verification
- Emergency SOS with preconfigured contacts
- Remote data wipe for sensitive information
- Live location sharing when internet resumes
- Encrypted SMS verification and two-factor command authorization

 Future work will prioritize privacy-preserving features and secure command authentication to prevent abuse.

Conclusion & Next Steps

MobiScan via SMS delivers practical mobile security and recovery tools that function without continuous internet access. By leveraging Android device admin APIs, SMS transport, and Firebase logging, it offers a lightweight yet robust solution for lost-or-stolen scenarios.

Thank youuuu!!!!!!

